Tutorial 3

Monday, March 10, 2025 07:08 PM

Chomsky Normal Form

A → ␣␣ BC / AD
B → ␣ a / b   ab x

S -> ASA ② | aB ① NT.T
A -> B ④ | S ⑤
B -> b | epsilon ③

① Start state S_0

$S_0 \rightarrow S$

② remove epsilons

$S_0 \rightarrow S$
$A \rightarrow B \mid S \mid \varepsilon$
$S \rightarrow ASA \mid aB \mid a$
$B \rightarrow b$

second round

$S_0 \rightarrow S$
$A \rightarrow B \mid S$
$S \rightarrow ASA \mid aB \mid a \mid SA \mid AS \mid S$
$B \rightarrow b$

③ break the more-than-2 NT pairs

$S_0 \rightarrow S$
$A \rightarrow B \mid S$
$S \rightarrow AC \mid aB \mid a \mid SA \mid AS \mid S$
$C \rightarrow SA$
$B \rightarrow b$

④ remove single ones (NT)

$S_0 \rightarrow AC \mid aB \mid a \mid SA \mid AS$
$A \rightarrow b \mid AC \mid aB \mid a \mid SA \mid AS$
$S \rightarrow AC \mid aB \mid a \mid SA \mid AS$

$A \to b \mid AC \mid aB \mid a \mid SA \mid AS$
$S \to AC \mid aB \mid a \mid SA \mid AS$
$C \to SA$
$B \to b$

5 change NT.T with an additional NT

$X \to a$
$S_0 \to AC \mid XB \mid a \mid SA \mid AS$
$A \to b \mid AC \mid XB \mid a \mid SA \mid AS$
$S \to AC \mid XB \mid a \mid SA \mid AS$
$C \to SA$
$B \to b$

NT → NT.NT
NT → T
no epsilon

Context Free Grammars

iii. Let $\Sigma = \{a, b\}$ and let $L = \{a^n b^m \mid m, n \in \mathbb{N}$ and $m \neq n\}$. Write a CFG for L.

$a^n b^m$ $\qquad m \neq n$ $\qquad 0^n 1^n$

regular ~~X~~

CFGs

$S \to aSb \mid \varepsilon$

$S \to aXb$
$X \to aX \mid Xb \mid \varepsilon$

aXb

$aX \mid a$ $\qquad Xb \mid b$

AX
$AaXb$
$AaaXbb$

Final Solution =
$S \to AX \mid XB$
$A \to aX \mid a$
$B \to Xb \mid b$
$X \to aXb$

~~AX~~
~~$a.\varepsilon$ = "a"~~
~~XB~~
~~$\varepsilon.b$ = "b"~~

$\cancel{\varepsilon} \cdot b$ = "b"

as its $\mathbb{N}$

a's are at most b's

at least as many a's as b's

$\underline{ab}$
b

$S \rightarrow \_\_\_\_$ ab | ba | b | ε

This is Ambigious

$a^n b^m$ $\underline{n - m - 5n}$ $n, m \in \mathbb{N}$

$S \rightarrow aXb \mid aXbb \mid aXbbb \mid aXbbbb \mid aXbbbbb \mid \frac{ab}{\varepsilon}$

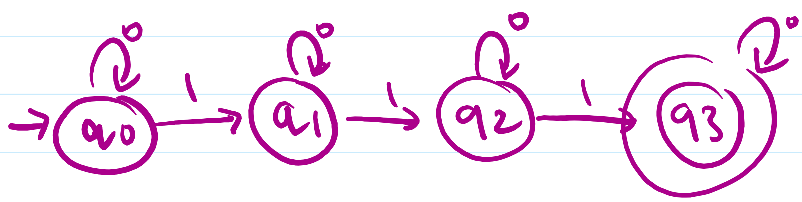
ε
ab

aXb
aaXbbb

aXbb
aaXbbb

A a. $\{w \mid w$ contains at least three 1s$\}$

111



$S \rightarrow XRXRXRX$

$R \rightarrow 1$

$X \rightarrow 0 \mid \varepsilon \mid XX \mid 1$

e. $\{w \mid w = w^{\mathcal{R}}$, that is, w is a palindrome$\}$

regular

1111

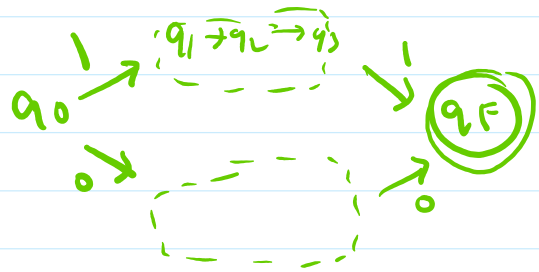
0110
0110

01
10 X

regular

$\longrightarrow$ $\longleftarrow$

$$\frac{0110}{0110} \qquad \frac{10}{10}\ X$$



$w = w^R$

$S \to 0S0 \mid 1S1 \mid \varepsilon$

0110
0S0
00S00
001100

$S \to X$
$X \to 0X0 \mid 1X1$

^{A}**c.** $\{w\#x \mid w^{\mathcal{R}}$ is a substring of x for $w, x \in \{0,1\}^*\}$

$\underbrace{0110}_{w}\ \#\ \underbrace{1\overbrace{1011}^{w}000}_{x}$

$S \to 0S0R \mid 1S1R \mid X$
$X \to \#$
$R \to 0 \mid 1 \mid RR \mid \varepsilon$

0S0R
01S10R
011S110R
011#110R
011#110RR
RRR
011#110111

***2.22** Let $C = \{x\#y \mid x, y \in \{0,1\}^*$ and $x \neq y\}$. Show that C is a context-free language.

$S \to \# \mid 1S0 \mid 0S1 \mid A \mid B$
diff y $\leftarrow A \to 0A0 \mid \#0 \mid \#1 \mid 1A1 \mid 0A1 \mid 1A0$
diff x $\leftarrow B \to 1B1 \mid 0\# \mid 1\# \mid 0B0 \mid 0B1 \mid 1B1$

diffx ← B → |B|| 0#||# |0B0 | 0B||1B|